M201103W- Signals and Systems

ssch7_5 –Inverse unilateral z-transform

14:10 pm Wednesday @ C107, 1st Teaching Building

14:10 pm Friday @ A107, 1st Teaching Building

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Inverse unilateral z-transform

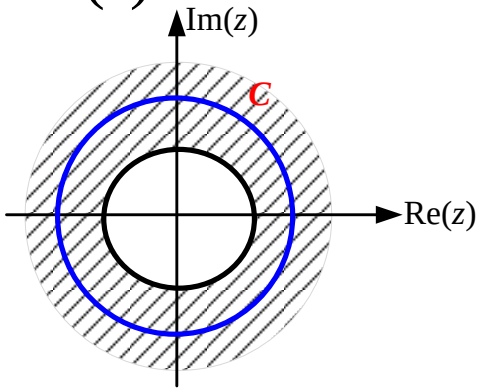
※ Inverse unilateral z-transform

$$x[k] = \frac{1}{2\pi \mathrm{j}} \oint_c X(z) z^{k-1} \mathrm{d}z$$

C is a closed curve in ROC of $X(z)$.

$$= \sum_i \mathrm{Res}\{X(z) z^{k-1}\}_{z=z_i}$$

z_i are the poles inside C of $X(z)z^{k-1}$.



Inverse unilateral z-transform

Several methods

※ Power series expansion

※ Residue method

Complicated to calculate, but used widely.

※ Partial fraction expansion method

Easy to calculate, but used only in rational fraction.

Power series expansion

※ Power series expansion：

$$X(z)=\sum_{k=0}^{\infty}x[k]z^{-k}=x[0]+x[1]z^{-1}+x[2]z^{-2}+\cdots$$

Expand $X(z)$ to power series by z^{-1} in the ROC, where the coefficients are $x[k]$.

$$x[k]=\{x[0],x[1],x[2],\cdots\}$$

Example. Calculate $x[k]$ using power series expansion, where $X(z)=\dfrac{2z^2-0.5z}{z^2-0.5z-0.5}$, $|z|>1$

Expand X(z) to power series by z^{-1} in the ROC.

$$
\begin{array}{r|l}
 & 2 \quad +0.5z^{-1}+1.25z^{-2} \quad +\cdots \\
\hline
z^2-0.5z-0.5 & 2z^2-0.5z \\
 & \underline{2z^2 \quad -z-1} \\
 & \quad 0.5z+1 \\
 & \quad \underline{0.5z-0.25-0.25z^{-1}} \\
 & \qquad 1.25+0.25z^{-1} \\
 & \qquad \underline{1.25-0.625z^{-1}-0.625z^{-2}} \\
 & \qquad\qquad \cdots
\end{array}
$$

Inverse z-Transform of $X(z)$: $x[k]=\{2,0.5,1.25,\cdots\}$

This method is simple, convenient and intuitive, but it is difficult to form a closed solution. It is practical to calculate several values of $x[k]$ using this method.

Partial fraction expansion method

※ Partial fraction expansion method：

$$X(z)=\frac{B(z)}{A(z)}=\frac{b_m z^m+b_{m-1}z^{m-1}+\cdots+b_1 z+b_0}{z^n+a_{n-1}z^{n-1}+\cdots+a_1 z+a_0}$$

$X(z)$ can be split into partial fractions.

Several elementary z-transform pairs:

$$a^k u[k]\xleftrightarrow{\mathscr{Z}}\frac{1}{1-az^{-1}}\quad |z|>|a|$$

$$(k+1)a^k u[k]\xleftrightarrow{\mathscr{Z}}\frac{1}{(1-az^{-1})^2}\quad |z|>|a|$$

$$-a^k u[-k-1]\xleftrightarrow{\mathscr{Z}}\frac{1}{1-az^{-1}}\quad |z|<|a|$$

$$-(k+1)a^k u[-k-1]\xleftrightarrow{\mathscr{Z}}\frac{1}{(1-az^{-1})^2}\quad |z|<|a|$$

Example. Calculate the inverse z-transform according to partial fraction method. $X(z)=\dfrac{2z^2-0.5z}{z^2-0.5z-0.5}$, $|z|>1$

Solution: There are two poles $z_1=1$, $z_2=-0.5$.

Split $X(z)$,

$$\frac{X(z)}{z}=\frac{(2z-0.5)}{(z-1)(z+0.5)}=\frac{A}{z-1}+\frac{B}{z+0.5}$$

$$X(z)=\frac{Az}{z-1}+\frac{Bz}{z+0.5} \qquad x[k]=[A(1)^k+B(-0.5)^k]u[k]$$

$$A=(z-1)X(z)\big|_{z=1}=1 \qquad B=(z+0.5)X(z)\big|_{z=-0.5}=1$$

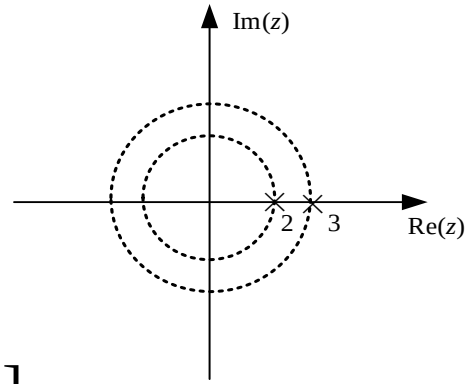
$$x[k]=[1+(-0.5)^k]u[k]$$

Example. Calculate inverse z-transform under different ROC conditions.

$$H(z) = \frac{1}{(1-2z^{-1})(1-3z^{-1})}$$

Solution: Rewrite:

$$H(z) = \frac{-2}{1-2z^{-1}} + \frac{3}{1-3z^{-1}}$$



(1) $|z| > 3$ $\quad h[k] = (-2^{k+1} + 3^{k+1})u[k]$

(2) $2 < |z| < 3$ $\quad h[k] = -2^{k+1}u[k] - 3^{k+1}u[-k-1]$

(3) $|z| < 2$ $\quad h[k] = 2^{k+1}u[-k-1] - 3^{k+1}u[-k-1]$